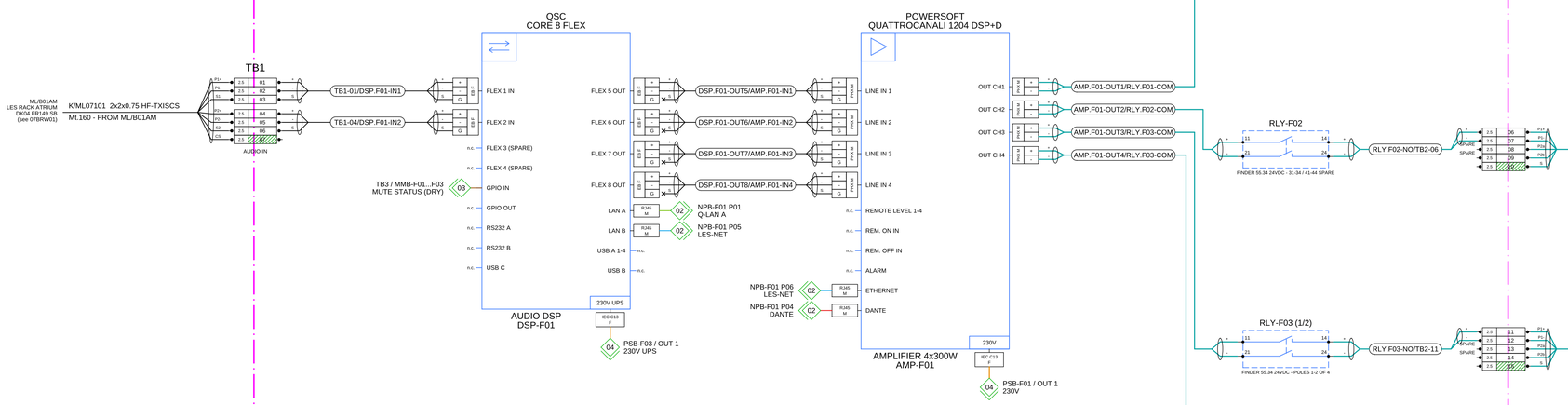
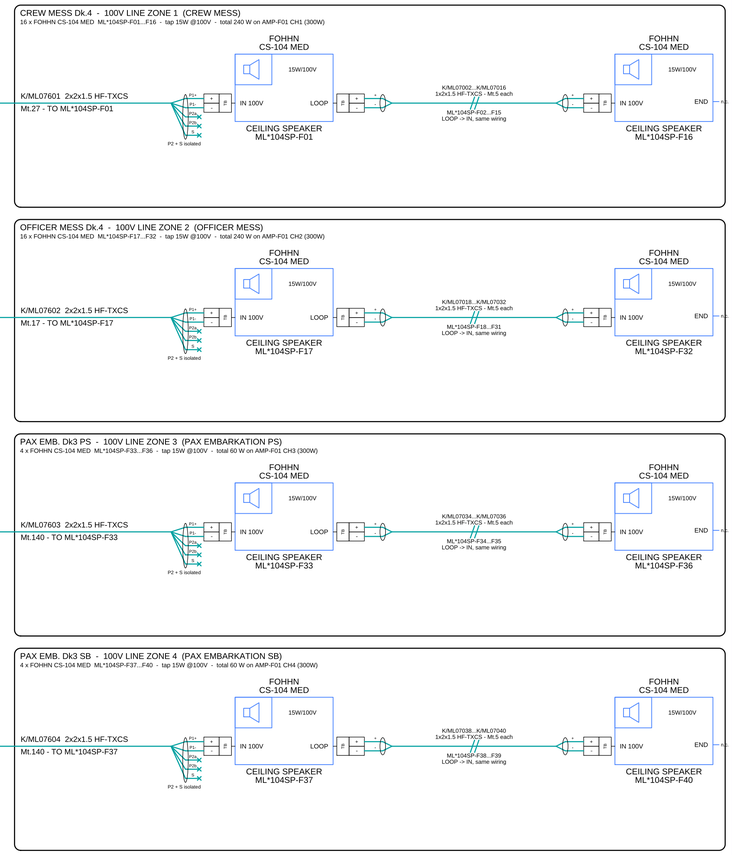
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SHEET 01 - AUDIO SIGNAL PATH / 100V SPEAKER LINES





AUDIO ZONE SUMMARY

ZONE	DSP-F01	AMP-F01	MUTING RELAY (COM-NO)	TB2	FIELD CABLE	LOAD @100V	MUTE BY
CREW MESS	FLEX 5 OUT	CH1	RLY-F01 11-14/21-24	TB2-01...05	K/ML07601	16 x 15W = 240W	MMB-F01
OFFICER MESS	FLEX 6 OUT	CH2	RLY-F02 11-14/21-24	TB2-06...10	K/ML07602	16 x 15W = 240W	MMB-F02
PAX EMBARKATION PS	FLEX 7 OUT	CH3	RLY-F03 11-14/21-24	TB2-11...15	K/ML07603	4 x 15W = 60W	MMB-F03
PAX EMBARKATION SB	FLEX 8 OUT	CH4	RLY-F03 31-34/41-44	TB2-16...20	K/ML07604	4 x 15W = 60W	MMB-F03

NOTE

1. Balanced audio: screen connected to the G pin of the RECEIVING device only; sending end cut back and taped.
2. Field cable screens earthed at rack side only (TB1-07 / TB2 PE terminals); field side cut back, isolated, taped.
3. 100V lines switched on both conductors by muting relays (NO contacts, coil energised = audio enabled).
   Relay coils, MMB boards and PAGA interface: see sheet 03. Relays: FINDER 55.34.9.024.0040 + socket 94.04.
4. DSP flex channels 1-2 configured as inputs, 5-8 as line outputs (+4 dBu) to AMP-F01 analogue inputs.
   Dante link AMP-F01 / Q-LAN on NPB-F01 P04 kept as per functional diagram (sheet 02).
5. AMP-F01 channels set to 100V constant voltage mode, high-pass 80 Hz (FOHHN CS-104 with 100V transformer).
6. Spare pair P2 of 2x2x1.5 home-run cables parked on TB2 SPARE terminals, isolated at speaker side.
7. Cable numbers K/MLxxxxx as per 006376A6F41026007; internal cables identified by FROM.DEV-OUT/TO.DEV-IN labels.

SYMBOLS

DEV.F01-OUT/DEV.F02-IN — internal cable label: FROM.DEVICE-OUTPUT/TO.DEVICE-INPUT

- cable connector: type / gender (M plug, F socket)
- multi-pin connector: pins drawn one by one
- cable sheath / screen (conductors fanned out)
- conductor cut back, isolated and taped (not connected)
- PE terminal (green/yellow, earthed on DIN rail)
- 02 — continues on sheet nn (number in diamond)
- device drawn in full on another sheet (partial view)

LINE TYPES

- BALANCED AUDIO (+ / - / S)
- SPEAKER LINE 100V
- Q-LAN (VLAN QLAN-A)
- LES-NET
- DANTE
- GPIO / CONTROL
- 230V AC
- RACK LIMIT

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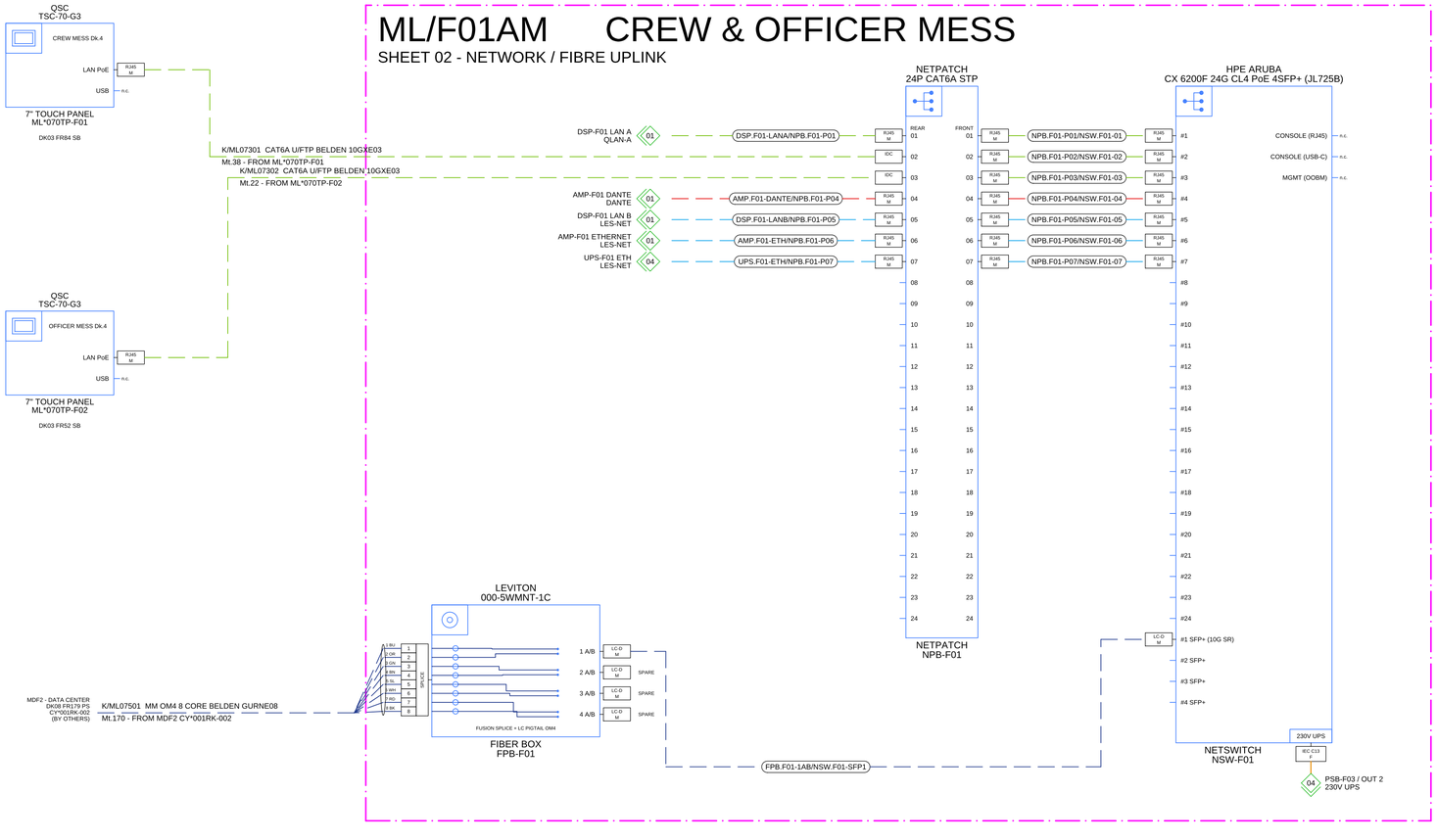
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SHEET 02 - NETWORK / FIBRE UPLINK



PORT ASSIGNMENT (ref. A6F41026007 sh.07)

NPB-F01	NSW-F01	DEVICE	PORT	VLAN	PoE	CABLE
P01	#1	DSP-F01	LAN A	QLAN-A	OFF	DSP.F01-LANA/NPB.F01-P01
P02	#2	ML*070TP-F01	LAN PoE	QLAN-A	802.3at	K/ML07301
P03	#3	ML*070TP-F02	LAN PoE	QLAN-A	802.3at	K/ML07302
P04	#4	AMP-F01	DANTE	DANTE	OFF	AMP.F01-DANTE/NPB.F01-P04
P05	#5	DSP-F01	LAN B	LES-NET	OFF	DSP.F01-LANB/NPB.F01-P05
P06	#6	AMP-F01	ETHERNET	LES-NET	OFF	AMP.F01-ETH/NPB.F01-P06
P07	#7	UPS-F01	ETH	LES-NET	OFF	UPS.F01-ETH/NPB.F01-P07
-	SFP+ #1	FPB-F01 LC1 A/B	UPLINK MDF2	TRUNK (ALL)	-	K/ML07501
P08-P24	#8-#24	NOT USED	-	-	-	-
-	SFP+ #2-#4	SPARE (FPB LC2-LC4)	-	-	-	-

VLAN COLOURS

QLAN-A — DANTE — LES-NET — UPLINK

DETAIL A - CAT6A U/FTP TERMINATION T568B (K/ML07301, K/ML07302)



T568B

Pair 1: pins 4-5 BU / WH-BU Pair 2: pins 1-2 WH-OR / OR
Pair 3: pins 3-6 WH-GN / GN Pair 4: pins 7-8 WH-BN / BN
Drain = foil bonded to keystone / plug metal shell (360 deg).
Max untwist 6 mm. Test: Cat6A permanent link (TIA-568.2-D), report required.

NOTE

1. Field Cat6A cables punched down on shielded IDC keystones (NPB-F01 rear); in-rack devices patched with factory CAT6A S/FTP LSZH cords (T568B straight). Front patch cords NPB-F01 -> NSW-F01 1:1.
2. VLANs as per A6F41026007 sheet 07: ports 1-3 QLAN-A, 4 DANTE, 5-7 LES-NET (access, untagged). SFP+ #1 uplink to MDF2: 802.1Q trunk, all VLANs tagged. IP plan / VLAN IDs by ship IT.
3. PoE 802.3at enabled on ports 2-3 only (QSC TSC-70-G3). Q-LAN QoS (DSCP) per QSC guidelines.
4. Fibre K/ML07501: 8 x OM4 fusion spliced to LC pigtails (TIA-598 colours); cores 1-2 -> LC1 A/B uplink, cores 3-8 spare on LC2-LC4. A/B polarity crossed on the LC duplex patch (A->B). OTDR test both ends.
5. Discrepancy found: A6F41026007 sh.06 shows UPS LAN on NPB-F01 P13; sheet 05 / 07 show P07 -> P07 used.
6. Device sheets: DSP-F01 / AMP-F01 = sheet 01, UPS-F01 = sheet 04, touch panels F03/F04 = 07BRW01.

SYMBOLS

DEV.F01-OUT/DEV.F02-IN — internal cable label: FROM.DEVICE-OUTPUT/TO.DEVICE-INPUT

RJ45 M — cable connector: type / gender (M plug, F socket)

multi-pin connector: pins drawn one by one

cable sheath / screen (conductors fanned out)

conductor cut back, isolated and taped (not connected)

PE terminal (green/yellow, earthed on DIN rail)

02 — continues on sheet nn (number in diamond)

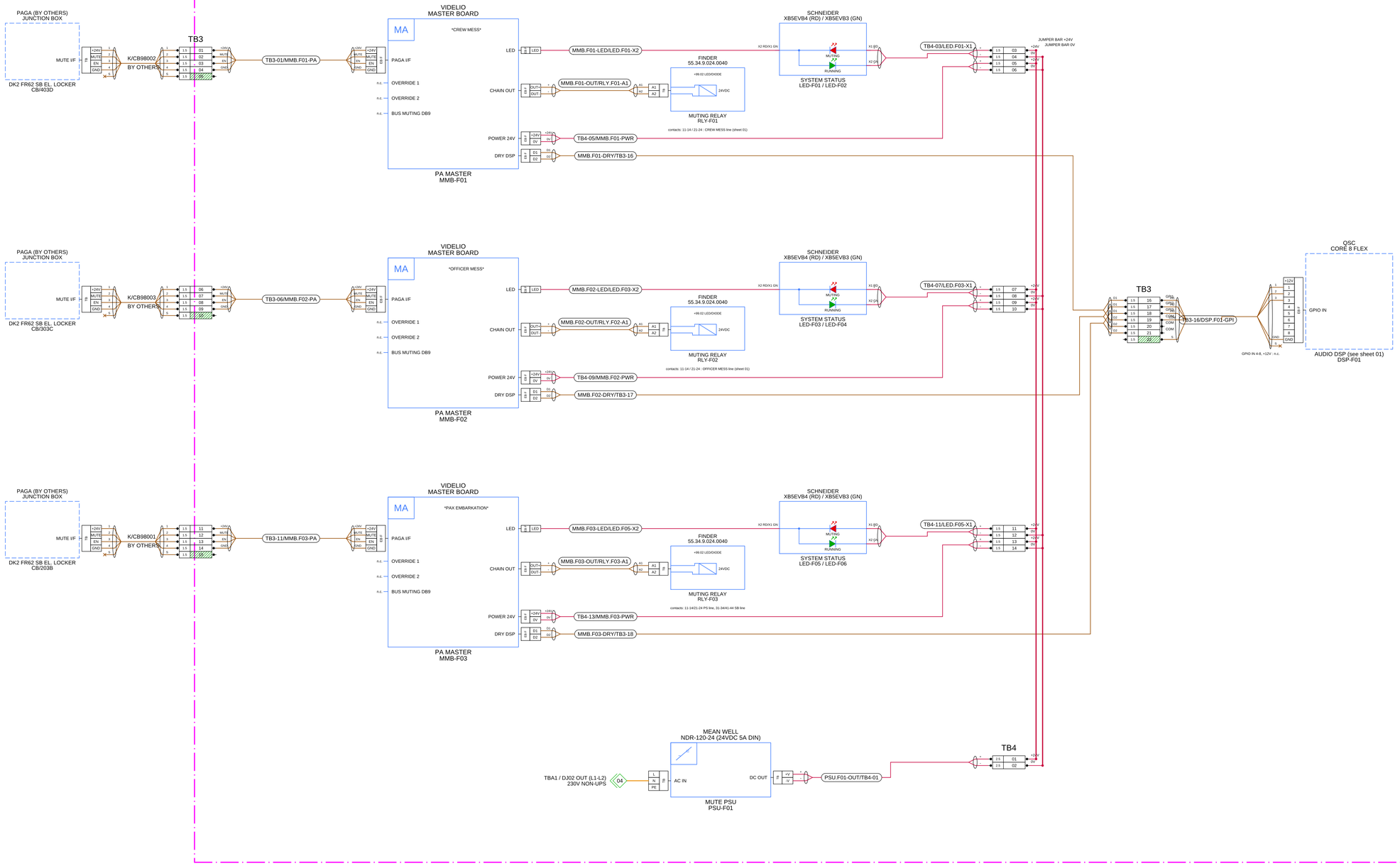
device drawn in full on another sheet (partial view)

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SHEET 03 - PAGA MUTING / STATUS LEDs / 24VDC / DSP GPIO



MUTING SUMMARY

BOARD	ZONE	PAGA JB	CABLE	TB3	RELAY	DSP	LEDs
MMB-F01	CREW MESS	CB/403D	K/CB98002	TB3-01...05	RLY-F01	GPI 1	LED-F01/LED-F02
MMB-F02	OFFICER MESS	CB/303C	K/CB98003	TB3-06...10	RLY-F02	GPI 2	LED-F03/LED-F04
MMB-F03	PAX EMBARKATION	CB/203B	K/CB98001	TB3-11...15	RLY-F03	GPI 3	LED-F05/LED-F06

NOTE

1. PAGA interface per A6F41026007 sh.04/06: MMB sends 24V TO PA. PA returns MUTE and ENABLE.
   Cables K/CB98001-003 and terminations at PAGA JB by PAGA supplier.
   JB <-> zone assignment (CB/403D, CB/303C, CB/203B) TO BE CONFIRMED with PAGA supplier.
2. Fail-safe: CHAIN OUT energises the muting relay coil when no PA call is active;
   PA call or MMB/PSU failure de-energises the coil and opens the 100V line (sheet 01).
3. MMB DRY DSP contact (closed = muted) reported to Q-SYS GPIO IN 1/2/3 for logic and fade.
4. SYSTEM STATUS lamps on rack front: RD = MUTING, GN = RUNNING (MMB LED output drives midpoint).
5. TB4: +24V terminals with fuse (2A MMB, 1A LEDs), 0V common; PSU-F01 fed from DJ02 (sheet 04).
6. Screens of PAGA cables earthed on TB3 PE terminals (rack side only).
7. Override 1/2 and BUS MUTING DB9 of MMB not used in this project.

SYMBOLS

DEV.F01-OUT/DEV.F02-IN — internal cable label: FROM.DEVICE-OUTPUT/TO.DEVICE-INPUT

- cable connector: type / gender (M plug, F socket)
- multi-pin connector: pins drawn one by one
- cable sheath / screen (conductors fanned out)
- conductor cut back, isolated and taped (not connected)
- PE terminal (green/yellow, earthed on DIN rail)
- 02 — continues on sheet nn (number in diamond)
- device drawn in full on another sheet (partial view)

LINE TYPES

- GPIO / CONTROL / PAGA I/F
- 24VDC
- 230V AC
- RACK LIMIT

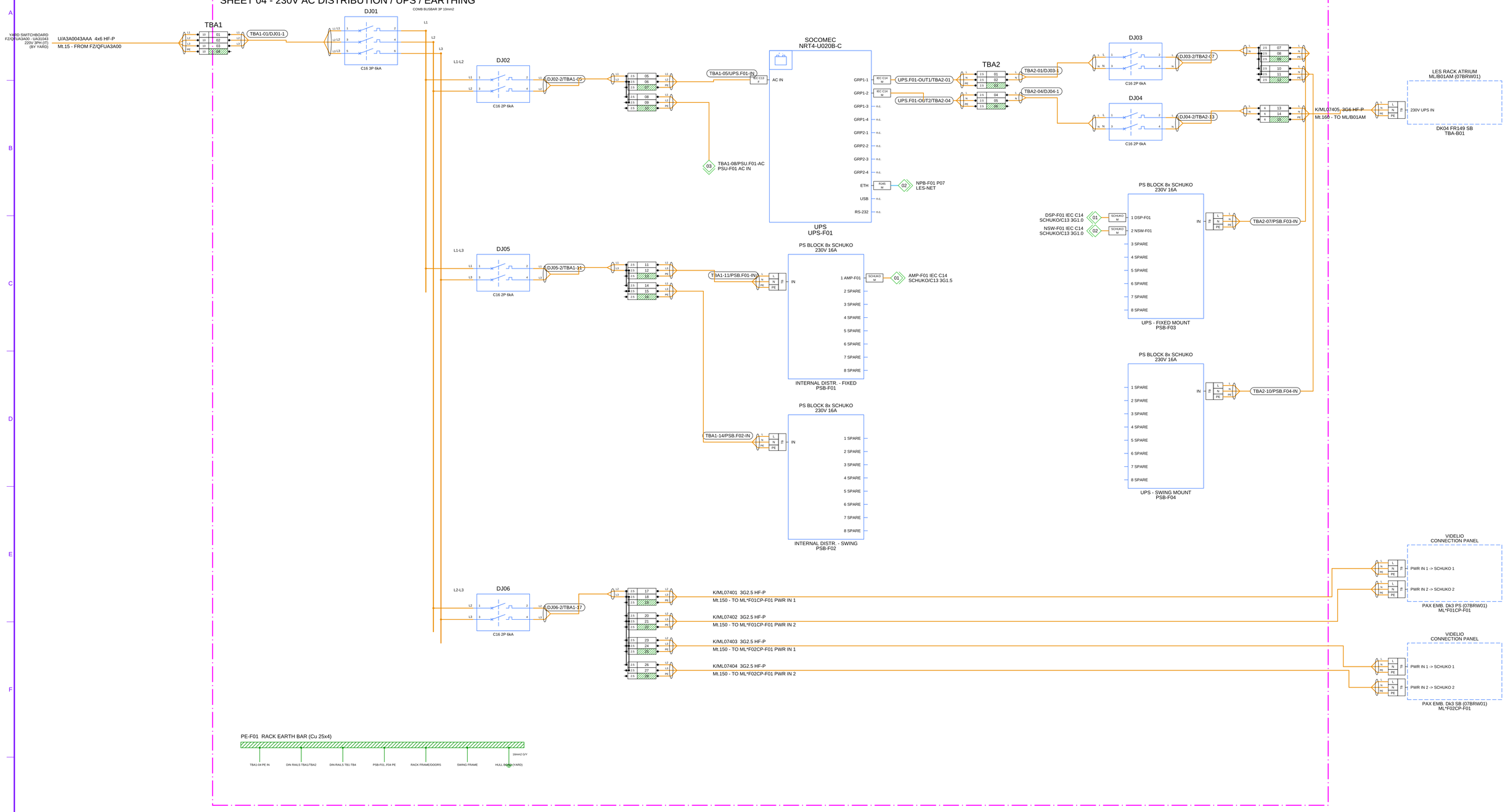
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SHEET 04 - 230V AC DISTRIBUTION / UPS / EARTHING



BREAKER SCHEDULE

MCB	RATING	SUPPLY	LOADS	
DJ01	C16 3P	L1 L2 L3	MAIN SWITCH	-
DJ02	C16 2P	L1-L2	UPS-F01 IN, PSU-F01 (mute)	non-UPS
DJ03	C16 2P	UPS GRP1-1	PSB-F03, PSB-F04	UPS
DJ04	C16 2P	UPS GRP1-2	K/ML07405 -> ML/B01AM	UPS
DJ05	C16 2P	L1-L3	PSB-F01, PSB-F02	non-UPS
DJ06	C16 2P	L2-L3	K/ML07401-404 -> CP F01/F02	non-UPS

NOTE

1. Ship supply 230V 3-phase IT (isolated neutral): single-phase loads connected line-to-line (L1-L2 / L1-L3 / L2-L3) as per A6F41026007 sh.06; all MCBs 2-pole (both lines switched).
2. A6F41026007 sh.05 shows panels fed from DJ05, sh.06 from DJ06: DJ06 (L2-L3) adopted. CONFIRM.
3. K/ML07401-404 3G2.5 Mt.150: verify voltage drop / Zs with actual connection-panel loads (16A not achievable at 4%).
4. UPS GRP1-1/1-2 IEC C13 outlets used with C14 cords to TBA2.
5. All PE terminals earthed on DIN rail to PE-F01 bar.
6. Internal wiring LSZH; ferrules on all stranded conductors.

LINE TYPES

- 230V AC
- PE / EARTH
- LES-NET
- RACK LIMIT

SYMBOLS

- DEV.F01-OUT/DEV.F02-IN — internal cable label: FROM.DEVICE-OUTPUT/TO.DEVICE-INPUT
- cable connector: type / gender (M plug, F socket)
- multi-pin connector: pins drawn one by one
- cable sheath / screen (conductors fanned out)
- conductor cut back, isolated and taped (not connected)
- PE terminal (green/yellow, earthed on DIN rail)
- 02 — continues on sheet nn (number in diamond)
- device drawn in full on another sheet (partial view)

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